BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: *[Home Security & Automation System]*
by

[Group_No-8]
[Section-02]

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Abstract

This project aims to create a multifunctional home security system and automation system to improve the safety and also energy efficiency.In this robotic system we have used Arduino Uno and ESP32 controllers with different sensors including RFID,PIR,Raindrop sensor,temperature sensor,flame sensor. RFID based door system enhance the home security,Flame sensor detect fire and send alert ,PIR sensor reduce energy wastes by controlling fan,raindrop sensors active servo motor to retrieve cloths from being wet.The methodology enclose a modular architecture,the system is programmed in C++ via Arduino IDE and the assimilation of IOT for remote alert system.Final result establish a promising autonomous operation of all main functionalities including proving a effective security,sending fire alert and promote to sustainable use of energy. .This home automation system robots impact mainly lies in cost effectiveness,scalable architecture ,and high potential to improve safety systems in smart homes.

Keywords

[3 - 5 keywords, "microcontroller, automation, IoT, sustainability, Iot,Security System ,sustainability "]

1. Introduction:

The increasing requirements for home security and sustainable energy consumption has facilitated the implementation of home security systems.This project is motivated by the need for affordable home security with simple integrated hardware and software systems.The primary objective of the project is to create a home security system that integrates different actuators with multiple sensors to enhance the security , environmental monitoring ,automated appliance control.In this system the scope comprise interfacing sensors and actuators with microcontrollers.A fundamental principle of robotics and computer interfacing is to create an intelligent system that is able to solve real problems by taking sensory inputs from the environment

2. Related Work/ Inspiration:

Home automation system is a common robotic system now a days but most often is it works in specific aspects like security door system These are typically designed in specific domain like climate or security or lighting .Combining Smart door with camera module in commercial and control climate or lightning creates complex expensive product which is not affordable for middle class people that motivates me to work in this project and make this system in affordable price with more functionalities by combining fire detection,rain sensing,energy efficient human detection and light \fan controlling with less cost.

From existing projects our system is different in holistic integration and multi functional cohesion .Instead of using separated devices for multiple functions we have merged the system with diverse tasks in a synchronous way . By using Cross functional logic the system is able to make decision depending on the sensors combining the functionalities of specific sensors like PIR sensor works for both light control and fan activation instead of only light control and the speed of the fan can be controlled by the temperature sensor.Unlike fragmented solution our project creates a single integrated system that perform multiple tasks at a time

3. Technical Approach

- ☐ **System Architecture:** .

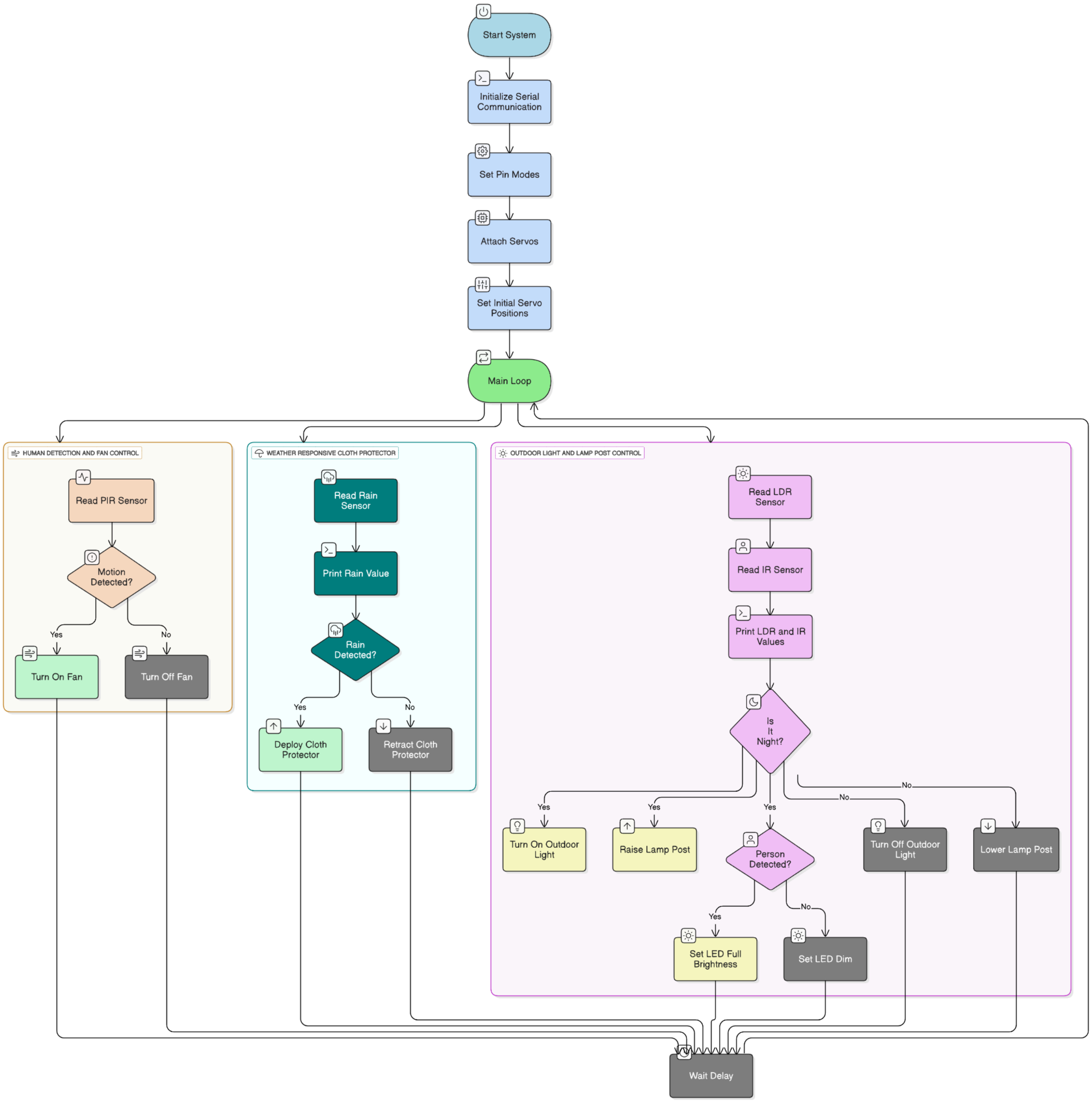
```
#include <Servo.h>

// --- Pin Configurations ---
```

```
Servo lampServo;        // Servo for lamp post control (Function
5)
Servo clothServo;       // Servo for cloth protection (Function 4)

int ledPin = 11;        // LED (outdoor light) pin (Function 5)
int ldrPin = A2;        // LDR pin (light detection for Function
5)
int irPin = 6;          // IR sensor pin for presence detection
(Function 5)
int servoPinLamp = 10; // Servo control pin for lamp post
(Function 5)
int servoPinCloth = 4; // Servo control pin for cloth protection
(Function 4)
int pirPin = 7;         // PIR sensor pin for motion detection
(Function 2)
int motorPin1 = 8;      // DC motor (fan) control pin (Function 2)
int motorPin2 = 9;      // DC motor (fan) control pin (Function 2)
int rainSensorPin = A1; // Rain sensor analog pin (Function 4)

// --- Constants ---
int ldrThreshold = 500;  // LDR threshold to decide day/night for
Function 5
int rainThreshold = 400; // Threshold for rain detection
(Function 4)
int pirThreshold = 500; // PIR sensor detection threshold
(Function 2)

void setup() {
  Serial.begin(9600);

  // Initialize pins for Function 5
  pinMode(ledPin, OUTPUT);    // LED (outdoor light)
  pinMode(ldrPin, INPUT);     // LDR sensor
  pinMode(irPin, INPUT);      // IR sensor for presence detection
  pinMode(pirPin, INPUT);     // PIR sensor for motion detection
  pinMode(motorPin1, OUTPUT); // DC motor control
  pinMode(motorPin2, OUTPUT); // DC motor control
  pinMode(rainSensorPin, INPUT); // Rain sensor (analog)

  // Attach servos to their respective pins
  lampServo.attach(servoPinLamp);  // Lamp post control (Function
5)
  clothServo.attach(servoPinCloth); // Cloth protector control
(Function 4)

  // Initial positions for servos
  lampServo.write(0);    // Lamp post down (Function 5)
  clothServo.write(0);    // Cloth protector down (Function 4)
}

void loop() {
```

```
  // --- Function 2: Human Detection and Fan Control ---
  int pirValue = digitalRead(pirPin); // Read PIR sensor for
motion detection
  if (pirValue == HIGH) {
    // Turn on the fan if motion is detected
    digitalWrite(motorPin1, HIGH);
    digitalWrite(motorPin2, LOW);
  } else {
    // Turn off the fan if no motion is detected
    digitalWrite(motorPin1, LOW);
    digitalWrite(motorPin2, LOW);
  }

  // --- Function 4: Weather-Responsive Cloth Protector ---
  int rainValue = analogRead(rainSensorPin); // Read rain sensor
value
  Serial.print("Rain Sensor: ");
  Serial.println(rainValue); // Print rain sensor value for
debugging

  if (rainValue < rainThreshold) {
    // Protect the clothes if rain is detected
    clothServo.write(90);  // Move servo to protect clothes
  } else {
    // Move servo back if no rain is detected
    clothServo.write(0);
  }

  // --- Function 5: Outdoor Light Control ---
  int ldrValue = analogRead(ldrPin);  // Read LDR value for light
detection
  int irValue = digitalRead(irPin);   // Read IR sensor value for
presence detection

  Serial.print("LDR Value: ");
  Serial.print(ldrValue);
  Serial.print(" | IR Value: ");
  Serial.println(irValue);

  if (ldrValue > ldrThreshold) {  // If it's dark (nighttime)
    digitalWrite(ledPin, HIGH);  // Turn on the LED (outdoor
light)
    lampServo.write(90);  // Raise the lamp post

    if (irValue == LOW) {  // If IR detects a person (presence)
      analogWrite(ledPin, 255);  // Set LED to full brightness
    } else {
      analogWrite(ledPin, 50);   // Dim the LED if no person is
detected
    }
  } else {  // If it's bright (daytime)
```

```
    digitalWrite(ledPin, LOW);  // Turn off the LED (outdoor
light)
    lampServo.write(0);  // Lower the lamp post
  }

  delay(100);  // Small delay to stabilize the loop
}
```

Workflow (Functionality 2,4,5)



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Esp code

```
#include <SPI.h>
#include <MFRC522.h>
#include <ESP32Servo.h>
#include <WiFi.h>
```

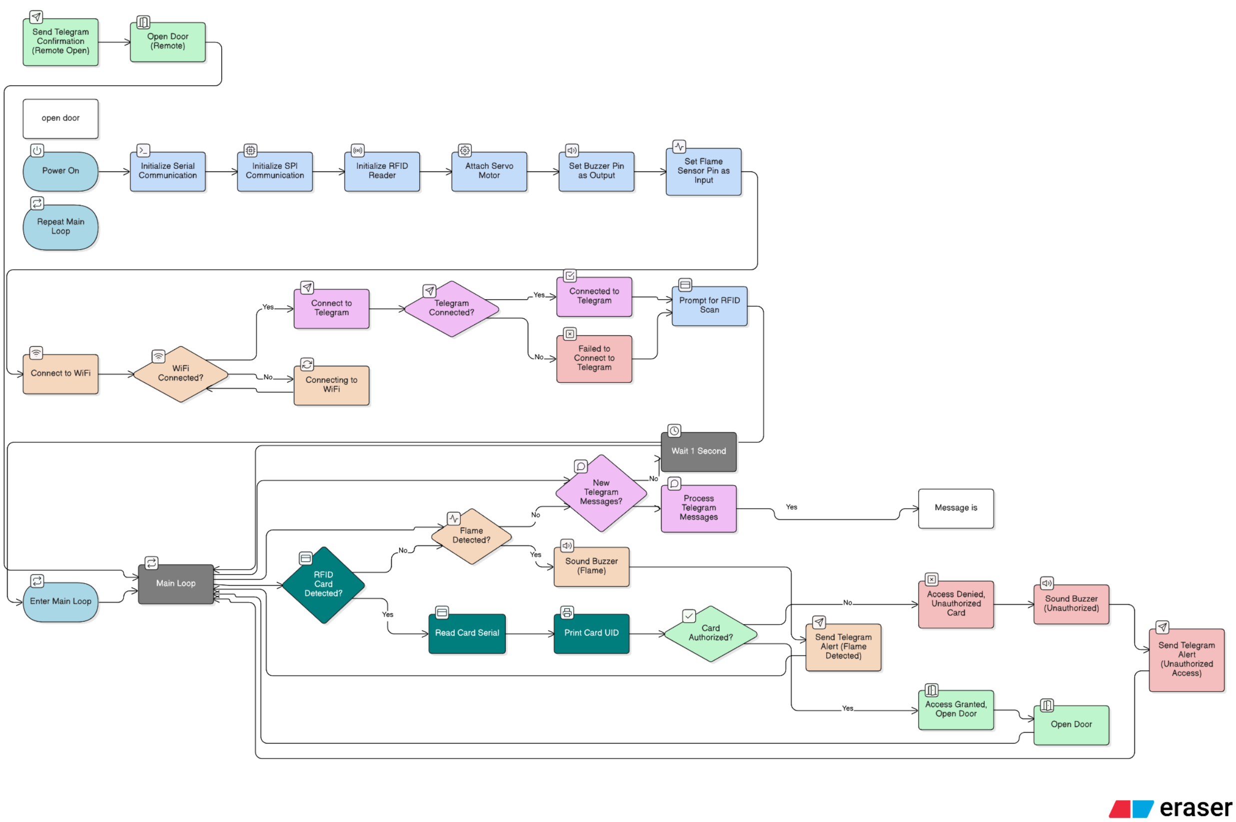
```
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>

// WiFi credentials
const char* ssid = "pial";          // Your WiFi SSID
const char* password = "12345678"; // Your WiFi password

// Telegram Bot Token (replace with your bot's token)
#define BOT_TOKEN "8290366140:AAE6LAtfoH5CL8F1atJ9mT0Ox-EzDIIIddk"

// Chat ID (replace with your chat ID)
#define CHAT_ID "1945561468"

// Pin Definitions
#define SS_PIN    5   // SDA (Slave Select) for RC522
#define RST_PIN   22  // Reset pin for RC522
#define SERVO_PIN 14  // Servo motor signal pin
#define BUZZER_PIN 15 // Buzzer pin
#define FLAME_SENSOR_PIN 34 // Flame sensor digital pin (DO)

// Create instances for MFRC522 and Servo
MFRC522 mfrc522(SS_PIN, RST_PIN);  // Create MFRC522 instance
Servo myServo;                     // Create Servo instance

// Flag to check if the door is open
bool doorOpen = false;

// Authorized RFID card UID (replace with your actual UID or expand
as needed)
byte authorizedUIDs[][4] = {
  {0x86, 0xEE, 0xB2, 0x1}  // Example authorized UID
  // Add more UIDs as needed
};

// Setup WiFi and Telegram
WiFiClientSecure client;
UniversalTelegramBot bot(BOT_TOKEN, client);

void setup() {
  Serial.begin(115200);    // Start serial communication
  SPI.begin();             // Start SPI communication
  mfrc522.PCD_Init();      // Initialize MFRC522
  myServo.attach(SERVO_PIN);  // Attach the servo to the pin
  pinMode(BUZZER_PIN, OUTPUT);  // Set the Buzzer pin as output
  pinMode(FLAME_SENSOR_PIN, INPUT);  // Set the Flame sensor pin as
input

  // Connect to WiFi
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(1000);
```

```
    Serial.println("Connecting to WiFi...");
  }
  Serial.println("Connected to WiFi");

  // Connect to Telegram
  client.setInsecure();  // Disable SSL verification if necessary
  if (client.connect("api.telegram.org", 443)) {
    Serial.println("Connected to Telegram!");
  } else {
    Serial.println("Failed to connect to Telegram.");
  }

  Serial.println("Scan your RFID card...");
}

void loop() {
  // Check if an RFID card is detected
  if (mfrc522.PICC_IsNewCardPresent()) {
    if (mfrc522.PICC_ReadCardSerial()) {
      Serial.print("Card UID: ");
      for (byte i = 0; i < mfrc522.uid.size; i++) {
        Serial.print(mfrc522.uid.uidByte[i], DEC);
        Serial.print(" ");
      }
      Serial.println();

      // Check if the card is authorized
      if (isAuthorized(mfrc522.uid.uidByte)) {
        Serial.println("Access Granted! Opening door...");
        openDoor();  // Open the door
      } else {
        Serial.println("Unauthorized access attempt!");
        soundBuzzer();  // Trigger buzzer for unauthorized card
        sendTelegramMessage("Unauthorized access attempt
detected!");
      }
    }
  }

  // Flame sensor logic (Trigger buzzer if flame is detected)
  int flameDetected = digitalRead(FLAME_SENSOR_PIN);  // Read flame
sensor digital output
  if (flameDetected == LOW) {  // Flame detected
    Serial.println("Flame detected! Triggering buzzer...");
    soundBuzzer();  // Trigger buzzer for flame detection
    sendTelegramMessage("Flame detected! Immediate action
required.");
  }

  // Telegram command listening (Check for "open door" command)
```

```
  int numNewMessages = bot.getUpdates(bot.last_message_received +
1); // Get new messages
  while (numNewMessages) {
    for (int i = 0; i < numNewMessages; i++) {
      String message = bot.messages[i].text;
      if (message == "open door") {
        sendTelegramMessage("Opening the door remotely.");
        openDoor(); // Open the door remotely
      }
    }
    numNewMessages = bot.getUpdates(bot.last_message_received + 1);
// Check for updates again
  }

  delay(1000);  // Delay to prevent continuous checking
}

// Function to open the door
void openDoor() {
  if (!doorOpen) {
    myServo.write(90);  // Open door (adjust the angle to 90
degrees)
    doorOpen = true;
    delay(5000);  // Keep the door open for 5 seconds
    myServo.write(0);  // Close door (adjust the angle back to 0
degrees)
    doorOpen = false;
  }
}

// Function to sound the buzzer for 1 second
void soundBuzzer() {
  digitalWrite(BUZZER_PIN, HIGH);  // Turn on buzzer
  delay(1000);                     // Keep buzzer on for 1 second
  digitalWrite(BUZZER_PIN, LOW);   // Turn off buzzer
}

// Check if the card is authorized (compare the UID)
bool isAuthorized(byte* uid) {
  for (int i = 0; i < sizeof(authorizedUIDs) / 4; i++) {
    bool match = true;
    for (byte j = 0; j < 4; j++) {
      if (uid[j] != authorizedUIDs[i][j]) {
        match = false;
        break;
      }
    }
    if (match) return true;
  }
  return false;
}
```

```
// Function to send a message via Telegram
void sendTelegramMessage(String message) {
  if (bot.sendMessage(CHAT_ID, message, "")) {
    Serial.println("Message sent to Telegram!");
  } else {
    Serial.println("Failed to send message.");
  }
}
```

Workflow(Functionality 1 and 3)



☐ **Components Used:**

Hardware

- Arduino Uno: Arduino Uno is the main microcontroller of our project that is used to control the actuators and the sensor.It works based on the input signals and after processing the output signals send to the motors
- ESP32 cam Module:Another controller that is working on wifi and camera for the alert system,sending pictures to the mobile through the wifi protocols when fire is detected and an unauthorized person wants to enter.
- Servo Motor: Servo motor works on PWM method,It is used to rotate in an angle.It Controls door lock & cloth rod movement.
- DC Motor: DC motor mainly converts electrical energy into mechanical rotation which is used in the fan system, It activates the fan from getting the signal from PIR and temperature sensors.Here DC motor is controlled by motor driver L298N
- RFID Sensor: Used for door access authentication.It is used radio frequency.
- PIR Sensor:Used to detect human movement by infrared radiation emitted from human body and control fan light activation system
  Flame Sensor: Detects fire hazards and IR light increase and turns the buzzer on and triggers ESP alert .
- Temperature Sensor: Measure room temperature and convert them into electrical signals,Arduino compares the input signal with the threshold and activates the fan .
- Raindrop Sensor: sense the rainfall and take steps for cloth protection.
- LDR Sensor: Measures ambient light ,In dark places, the resistance is high which creates low output voltage and in bright the resistance is low, which creates the high output voltage for smart lighting.
- Buzzer: Arduino controls the buzzer by sending high signal to activate the alerts on unauthorized access and fire.
- Breadboard, Jumper Wires, Resistors, Battery,multimeter etc are the some basic components of the projects.

Software

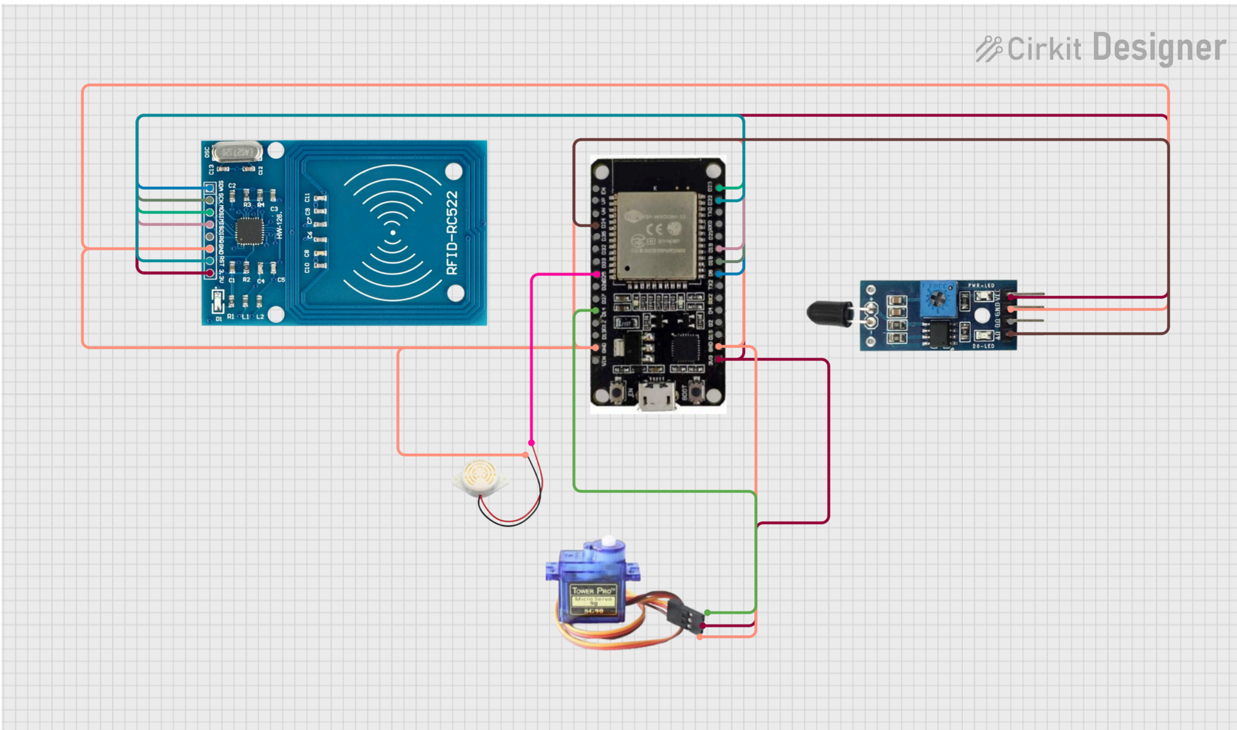
- Embedded C / Arduino C++: Programming language.
- ESP32 cam Libraries (Wi-Fi, Blynk): For notification system.
- Tinkercad: For circuit design & simulation.

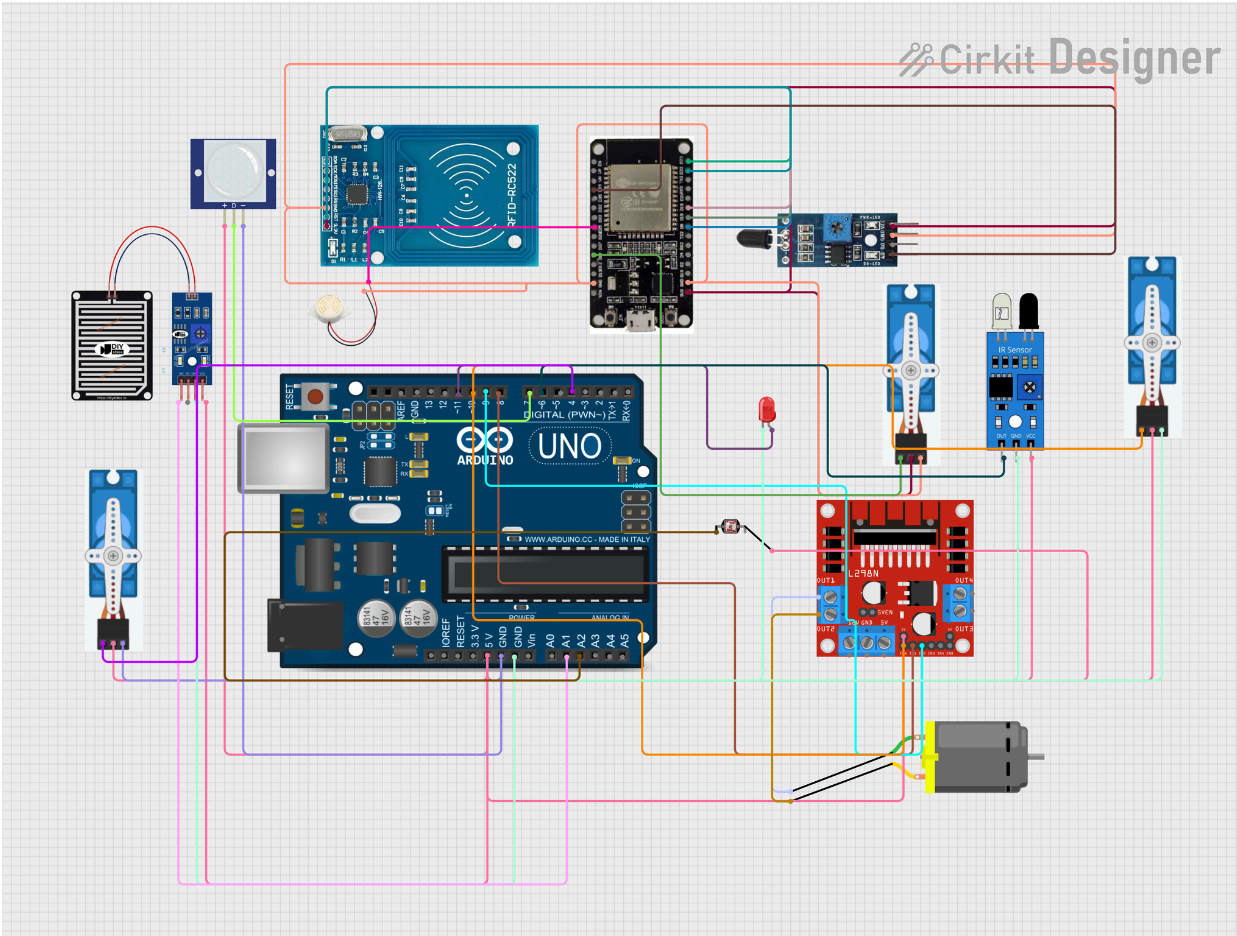
☐ **Cost Breakdown:** We have tried to complete this project in minimum cost

No	Components	Quantity	Unit Cost (BDT)	Total Cost (BDT)
1	Arduino Uno	1	1000	1000
2	RC522 RFID card reader module kit	1	200	200
3	PIR motion sensor	2	110	220
4	Temperature sensor	1	200	200
5	Flame Sensor	1	70	70
6	L298N H- Bridge dual motor driver	1	195	195
7	ESP32 module	1	450	450
8	3V DC Hobby motor with fan	1	120	120
9	Servo Motor	1	150	150
10	Buzzer	2	20	40
11	Jumper wire set	40	100	100
13	Breadboard	2	150	300
14	Battery	2	74	148
15	Dc motor	1	365	365

16	PWM	1	250	250
Total Cost (BDT)				3830

- [ ] **Circuit/Schematic:** Provide diagrams.





☐ **Functionality:**

In our home security system project ,we are working on 5 functionality.The working procedure is given below.

Step 1:The RFID sensor scans the RFID card and if the card is authenticated the servo motor will open the door for the person .If not ,an alert will be sent to the owner of the house and if the owner wants to enter the person ,the owner will send “open door” message and servo will open the door .

Step2: After entering the house ,the PIR sensor detect the human and turn the fan on by using servo motor and if the person leave the place the fan will again turn off

Step3-If any unwanted fire occurs in the house,flame sensor detects the fire and buzzer will be activated and also a message will be sent to the registered number by esp module.

Step4-When rainfall starts,the rain sensor detects the rain and the servo motor rotates the cloth stick and protects the cloth from getting wet.

Step5-By using LDR sensors ambient light level is observed and when it's dark the system turns on the light and also if a human is detected the brightness of the light will increase.

Challenges: We have faced different types of challenges to make this project

Technical Challenges:

1.Component Change- We decided to use an ESP32 cam but it does not have enough ports so we decided to shift to ESP32.

2.Sensors Regulation-Different sensors has different sensitivity,it was difficult to merge them after testing

3.Power supply Issue-Arduino operates in 5V but ESP32 operates in 3.3v.These different voltage ranges also created issues.

Integration challenges:

1.Power failure-TThe charge of the 9v battery was finished multiple times which creates issue when using Dc motor

2.Temperature sensor fluctuation-It was very difficult to have a proper signal from the temperature sensor because of circuit noise.

Budget Limitation: We are supposed to do an impactful robotic system at minimum cost.So we had to minimize many components like extra batteries,motors,breadboard etc.

4. Sustainability & Impact

- **Sustainability:**The project focus on energy efficiency by automatically control the fan and light when no human available in the room that save electricity.Using of dc motor efficient than using AC fan.In this project multiple components is used which are useable
- **Impact:**This project has strong impact on users for increasing the safety and immediate alert after detecting fire which make user more reliable on this system.Moreover automating tasks helps the elder people .Finally its affordable price make people achieve this system without hampering their daily life
- **Future Work:**This project can be more functional in future by including mobile app for control,solar power integration,adding more sensors and include machine learning models on sensor to predict behavior
- **Limitations:**The short range of RFID,setup of Arduino Uno which has some restriction in adding components,limited wifi range of ESP32 are some limitations of this system were observed despite the successful implementation of the project

5. Results & Discussion

Testing outcomes:

1 RFID sensor

Test	Expected Outcome	Actual Outcome	Result

1.Authorized card	Servo open the door	Door open Successfully	Correct
2.Unauthorized card	RFID send alert	Send alert successfully -Remaining the door close(No message form the owner)	Correct
3,Unauthorized card	RFID send alert	Send alert successfully -Open the door(If “Open Door” message is send from the owner)	Correct
3. NO card	No action	No action	Correct

2.Flame sensor

Test	Expected outcome	Actual outcome	duration
1.Flame present	Flame detect and send alert	Flame detect and send alert	5s
2.No flame	No response	No response	

3.Rain sensor

Test	Expected outcome	Actual Outcome	Result
1.Rain	Servo rotates	Servo rotates	Protect cloth
2.No rain	No response	No response	Keep as it is

4.PIR sensor

Test	Expected output	Actual output
Motion detect	Servo control fan	Servo response
No motion detect	No response	No reason

Result : All the functionalities performed perfectly.Unauthorized card is detected and send alert,fire alert is send , fan control is working, light automation is controlled,and rain drop sensor is also responding.Overall the project is reliable for people and it takes vary less time to response .The efficiency of this project is high for giving appropriate result for every sensors .

Discussion: The home security and automation system merged multiple sensors and actuators for creating a smart home.This system confirms the authorized locked system,fire alert system,light and fan control and responsive to the weather.The fast response, make the project more effective.The overall system ensure to use less energy.

6. Conclusion

This project prosperously implemented a cost effective,energy efficient, integrated Home security and automation system.The key achievements of this project is accumulation of different actuators with multiple sensors for establishing a system that manages access control,risk detection,climate control etc.This system proves a small computer can be used to create a safe home using energy efficient tools with less cost.The outcomes ensures that it is a economical, practical ,efficient,effective embedded system that protects dwelling,save energy ,enrich daily activities .

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
A. K. M Foyshal Sarkar	22201686	Example: Hardware design,software design	RFID door sensor using servo motor and send alert
MD. Shourove Shahriar	22201474	Example: Software Design,hardware design	Using Pir Sensor controlling the fan

Sabeha Farid	22101790	Example:software,hardware	Fire detect by flame sensor and send alert
Tayeba Tabassum Mridula	2201275	Example:software,hardware	Rain fall detect and rotate stick
Ramisa Jannat Chowdhury	22201288	Example:software,hardware	Using LDR sensor controlling the light

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